

Pointers & File Lab questions

61. Define a structure inventory containing item name, item number, and price.

Create an array of structures for n products and use a pointer to structure to:

- A. Read data for all products
- B. Display the data using pointer arithmetic
- C. Demonstrate that `ptr->member` and `(*ptr).member` give the same result

62. Write a C program that uses a pointer to structure to search for a product whose price is greater than a given value.

The program should:

- A. Use pointer increment (`ptr++`)
- B. Display all matching records
- C. Count and display the number of such products

63. Given a structure containing an integer and a pointer member:

```
struct sample {  
    int count;  
    int *p;  
};
```

Write a program to demonstrate and explain the effect of the following expressions using proper output:

- A. `++ptr->count`
- B. `(*ptr->p)++`
- C. `ptr++->p`

Also explain operator precedence based on program behavior.

64. Write a C program to create a file named student.txt and store roll number, name, and marks of n students using file I/O functions. Display the stored data by reading it back from the file.

65. Write a program to:

- A. Open an existing file in read mode
- B. Count the number of characters, words, and lines
- C. Display the count on the screen

(Use character-by-character file reading)

66. Write a C program to copy the contents of one file into another file. Your program should:

- A. Open source file in read mode
- B. Open destination file in write mode
- C. Handle the case when the source file does not exist

67. Create a file numbers.txt containing integers. Write a program to:

- A. Read all numbers from the file
- B. Find and display sum, average, maximum, and minimum

68. Write a menu-driven program using files that allows the user to:

- A. Add records to a file
- B. Display all records
- C. Append new records
- D. Exit

(Use appropriate file opening modes: w, r, a)

69. Write a C program to demonstrate the role of buffering in file handling. Show, using output and explanation:

- A. What happens when data is written but file is not closed
- B. How `fclose()` ensures data is flushed to disk

70. Write a program that opens two files simultaneously:

- A. One for reading
- B. One for writing

Copy only those lines from the first file that contain a specific word (entered by user) into the second file.